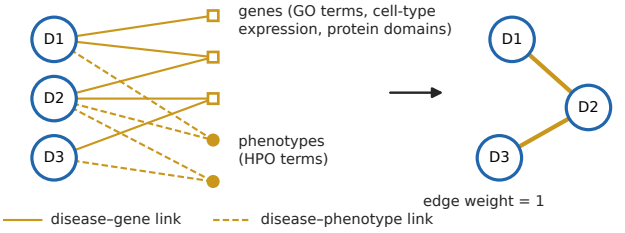
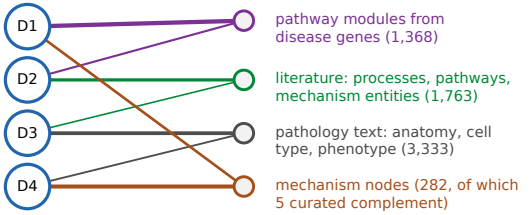
a Broad channel: shared genes and phenotypes



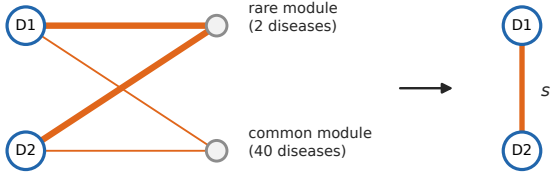
25,972 disease–gene links (11,096 diseases) and
244,394 disease–phenotype links (10,307 diseases).
Four similarity views (GO, phenotype, cell-type expression, protein domain) are rank-aggregated; the best pairs in each evidence class, and at least 3 per disease, are kept:
332,082 edges on 13,021 diseases as built; 848 of these are later removed as FDA control pairs (see d).

b Specific channel: shared mechanism modules



422,804 disease–module links (5,198 diseases;
6,746 modules, each shared by at least two diseases).
Link weight e = number of independent source families + ln(1 + dated documents). A link is kept if at least two source families or two dated documents support it.
Line width = e.

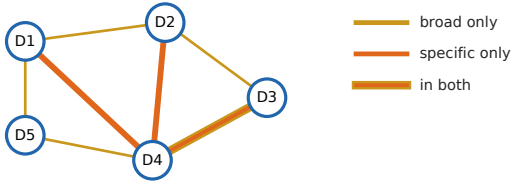
c Specific channel: projection to disease–disease

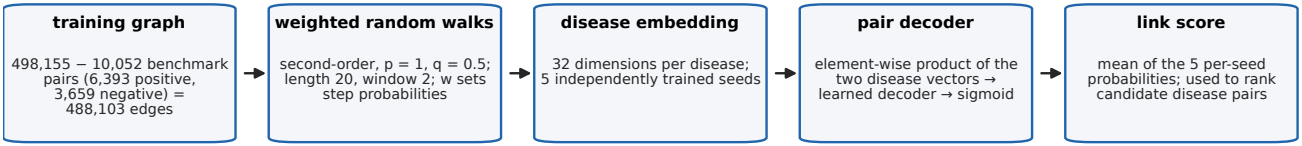


$x_{dm} = \sqrt{e_{dm} / (\ln(1 + n_m)\, L_d)}$, $s_{ij} = \sum_m x_{im}\, x_{jm}$

n_m = diseases on module m; L_d = modules on disease d.
Rare shared modules count more. Each disease keeps its 50 strongest partners: 191,298 edges; 191,019 on 5,194 diseases after removing 279 that reach a disease outside the broad graph.

d Combined graph and walk weights



$w = 1[\text{broad}] + 2\, f\, s / \bar{s}\; 1[\text{specific}]$
499,359 edges; removing 1,204 FDA repurposing control pairs leaves 498,155 on 13,021 diseases (331,234 broad, 190,007 specific, 23,086 both); w from 0.07 to 79.1.
Control pairs removed: 848 broad + 1,012 specific, 656 in both channels.
Balancing f: of 33 mechanism classes, those with less walk weight than complement have their edges raised (at most ×10; 20,049 edges raised); otherwise f = 1.

e From graph to embedding and link score